Technical Design Report

Project: Electrical Control Panel design for Industrial Machines

Origine : document de reference tiers anonymise

Date: 28-09-2025

Standard followed: IEC 60364

1. Executive Summary

This Report presents the design of an electrical control panel powering three machines industrial environment. The design includes load calculation, Cable sizing, selection of protection devices and compliance with IEC standards.

2. Scope & Assumptions

The scope covers the electrical design for

-Machine A: 5KW, 400V, 3-phase motor

-Machine B: 2KW, 400V, 3-phase motor

-Machine C: 2KW assumed, 230V, Single phase (lighting & sockets)

Assumptions:

-Power factor (pf) = 0.85 (motors)

-Motor Efficiency = 92

-Cable safety factor= 1.25

-Motor Starting current 6*Full load Current

-Ambient temperature= 30 C

-Installation method: Cables in trunking/ducting

-Protective devices selected according to IEC 60364

3. Load calculations

Machine A (5KW, 400V, 3-phase): IA 9.43 A

Machine B (2KW, 400V, 3-phase): IB 3.77 A

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Machine C (2 KW, 230V, Single-phase): IC 8.70 A

Design Current (with safety Factor 1.25):

-Machine A: 11.8 A

-Machine B: 4.7 A

-Machine C: 10.9 A

4. Protection Devices & Cable Sizing

Machine A (5 KW motor): 20 A MCCB+ Contactor+ Overload Relay, Cable: 2.5 mm^2 Cu

Machine B (2 KW motor): 10 A MCB + Contactor+ Overload Relay, Cable: 1.5 mm^2 Cu

Machine C (Lighting & Sockets): 16 A MCB + 30mA RCD, Cable: 2.5 mm^2 Cu

Incoming Supply: 63A, 4P MCCB (Main)

5. Distribution Board Layout

The DB consists of:

-Main isolator (63 A, 4P MCCB)

-Feeder 1, Machine A

-Feeder 2, Machine B

-Feeder 3, Machine C

6. Single Line Diagram (SLD)

A CAD wining diagram and SLD have been prepared (see attached Auto CAD files in repository).

7. Cable Schedule

From	To	Load	Cable Size	Length (m)	Protection	Remarks
DB	Machine A	11.8 A	2.5 mm^2	15	20 A MCB+OLR	3-Phase
DB	Machine B	4.7 A	1.5 mm^2	10	10 A MCB+OLR	3-Phase
DB	Machine C	10.9 A	2.5 mm^2	20	16 A MCB+RCD	230V, 1-phase

8. Safety & Standards Compliance

Design complies with IEC 6064. RCD protection included for sockets. Load balancing ensured. Emergency stop in the control circuit.

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9. Testing & Commissioning Plan

Insulation resistance, continuity, earth resistance, functional test, RCD test, overload relay test.

10. Conclusion

The proposed design ensures safe, reliable and standards compliant operation of the machines. Future recommendation integrates energy metering and remote monitoring.

11. Appendices

-Appendix A: Auto CAD wiring diagrams

-Appendix B: BOM (Excel)

-Appendix C: Standards references

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